

Phase 3 — AP / GL, Validation & Accruals Spec

PREPARED BY Energy Hippo implementation team

"Do actual billing" end to end and **produce real AP/GL interface files**. Lights up AccountsPayableDetail, APApprovalAndUpload, GLAccountUDV, BillValidation(+Status), OpenAccruals, and my_exports.

Entry path (see [entry-path-and-gaps.pdf](#)): bill entry, GL accounts, account AP/expense assignment, bill validation rules, and the AP **approval + upload run** are **UI-DIRECT/UI-THEN-RUN**. **Gaps**: there is **no UI to define a new AP/GL export format** (A1 — only the 2 built-ins export_ap/ export_ap2 are usable by hand; richer ERP formats are dev work); **accruals have no UI/API at all** (A2 — not hand-enterable); the **GL user-defined-fields view is not wired** (A3 — base-account coding only until finished). These are the phase's biggest reframes.

EEMSuite facts (verified): export eligibility gates on `BillingAccount.Type_Code` (0 No Upload, 1 AP, 2 GL, 3 AP+GL) + active status. GL accounts in `GLAccount` (`Type_Code` 0 Assets / 1 Expenses / 2 Liabilities) with 5 user-defined segments (`GLAccountUDField/GLAccountUDValue`). Bill lines carry GL via `BillItem.BI_ExpAcctID`. Export engine = `ExportSetup` (per-format Approval/ UploadFile/Batch procs) → `ExportFile/ExportLog`; `ExportType` rows APINV/APDIS. 47 `BillValidation` rules. Accruals = bills tagged commodity 2300 (Accruals) / 2301 (Reversals), paired in `Accrual_Bill/BillAccrualMap`.

Source-model extension

Add: `gl_account` (company, type 0/1/2, code, description), `gl_ud_field` (alias GLUDF1–5, name), `gl_ud_value` (`gl_account`, field, value), `ap_export_setup` (name, format, file_type, status_code, builder_ref), `bill_validation_rule` (test_id, measure_type, pct_threshold, abs_threshold, mode), `bill_validation_result`, `accrual` (`bill_ref`, `reversal_bill_ref`, commodity 2300/2301). Bill charge lines (Phase 1) gain `gl_account_code` population here.

1. GL chart of accounts

Per company, expense accounts by commodity + AP control:

code	type	description
5100-ELEC	1 Expenses	Electricity expense
5110-GAS	1 Expenses	Natural gas expense
5120-WATER	1 Expenses	Water/sewer/stormwater expense
5130-THERMAL	1 Expenses	Chilled water / steam allocation
5140-SOLAR	1 Expenses	Solar PPA expense
2000-AP	2 Liabilities	Accounts payable control

5 GL user-defined segments (GLAccountUDField, aliases GLUDF1–5 so the Sacramento-County-style exporter resolves them) — realistic university dimensions:

alias	segment
GLUDF1	Fund (e.g. 110 Operating, 250 Auxiliary, 700 Plant)
GLUDF2	Department (Facilities, Housing, Athletics, Labs)
GLUDF3	Program (Instruction, Research, Residential, Admin)
GLUDF4	Activity / Project
GLUDF5	Object code

gl_ud_value populates these per GL account with realistic + **partly-blank** coverage (to exercise empty-segment formatting). Building → Fund/Department mapping: campus buildings → Operating/Facilities; Cedar Row → Auxiliary/Housing; thermal plant → Plant.

2. Bill GL coding & account Type_Code

- Each Phase-1 bill_charge_line gets BI_ExpAcctID = the commodity's expense account; BillingAccount.BA_DefExpAcctID / BA_DefApAcctID set as fallback.
- Account Type_Code mix: most accounts **1 (AP)**; the campus electric + thermal accounts **3 (AP+GL)** (to exercise GL journal export); **one** Cedar Row account **0 (No Upload)** as the negative control (never exports).

3. AP/GL interface formats (produce one of each)

ap_export_setup rows → seven formats (builder procs supplied at load):

setup name	format	structure
export_ap	fixed-width .txt	single record/line (built-in)
export_ap2	.csv	fixed-width-in-CSV at bill grain (built-in)
nl_ap_gludf	fixed-width header+detail	GL-UDF coded (Sacramento-County style — region matches Northlake)
nl_ap_voucher	fixed-width C/H/L/D	PeopleSoft AP voucher
nl_gl_journal	fixed-width H/L	PeopleSoft GL journal, balanced debit/credit pairs
nl_sap_fi	comma-delimited	SAP FI document
nl_ar_interface	fixed-width	AR (tenant rebilling, Phase 4)

Each ExportSetup registers a SiEAppObjects EX object (export_ap/export_ap2 exist; the rest are new). A run writes one ExportFile + ExportLog rows and advances bill status to G/L Upload. SysConstants 'UploadDirAP' + AP_EMAILER_TASK required.

4. Bill validation

bill_validation_rule thresholds per measure type (modes 1 Bill Save + 2 Batch):

rule family	example threshold
cost change %	35% with \$1000 floor
cost max	\$1,000,000
usage change %	40%
usage floor	kWh 1000 / gas-water 100
period length	26–36 days
missing field / inactive account	presence checks

Fixtures: bills that **pass**; bills that **trip** each family (a cost spike, a short period, a usage zero); ≥ 1 **open** bill_validation_result **alert on an unapproved bill that blocks AP export**; a BillScenario 'Weather Normalized' for the two weather-normalized rules. Lights up rp_BillValidation (failures) + rp_BillValidationStatus (config matrix).

5. Accruals

- Set SysConstants 'MPA Accruals'.Value_I = 2300, 'MPA Reversals'.Value_I = 2301; ensure commodities 2300/2301 exist.
- Open accrual:** a Valley Electric account whose current period has no actual bill yet, with a manual bill carrying a 2300 line (no Accrual_Bill row) → appears in rp_OpenAccruals @Reverse=0.
- Trued-up pair:** run @Reverse=1 (or seed the +/- pair + Accrual_Bill link) so accrual(+) and reversal(-) net to zero when the actual arrives.

expectedAssertions

- gl_coding_present — every AP-eligible bill line resolves a GL account + UD segments; rp_GLAccountUDV returns the 5-segment grid.
- type_code_gating — the Type_Code=0 account never appears in an export; Type_Code 1/3 accounts do.
- one_file_per_format — running each ap_export_setup produces one ExportFile traceable to its bills; the GL journal balances (Σ debits = Σ credits).
- export_blocked_by_alert — the bill with an open validation alert is excluded from approval/upload.
- validation_pass_and_fail — rp_BillValidation shows the deliberate failures; clean bills pass to Status 2.
- open_accrual_then_reversal — rp_OpenAccruals lists the open accrual; after reversal the pair nets to 0.